

Interview Instructions

Estimated Time

This is an offline take-home interview. We recommend spending around 1 to 1.5 hours on it. Take the time you need — we're not tracking the exact duration.

Open-Book Format

This is an open-book challenge. You can use any tools you want, including documentation, online resources, and AI.

What to Submit

- Share a link to Amplitude event streaming webhook destination you created
- Send your code as an attachment. **Please make sure to remove any credentials like client secrets or API keys before submitting.** Share the file via Google Drive with private access or as an email attachment to Leo (leo.li@amplitude.com).
- Share a link to the Amplitude User Profile that includes the Hubspot source events you ingested
- Record a 5–10 minute Loom video explaining your solution. Please cover the following topics (and feel free to include anything else that helps us understand your approach):
 - Webhook:
 - Walk through your FTL code line-by-line — what is each part doing?
 - Show the results in the webhook destination.
 - What tools did you use to develop this? How did each help?
 - How did you approach debugging any issues?
 - If you used AI tools (e.g. ChatGPT, Copilot, Claude), walk through the actual prompts and conversations you had.
 - Ingestion:
 - Explain your ingestion logic/code.
 - Show results in the Amplitude User Profile.
 - What tools helped you here?
 - How did you debug ingestion issues?
 - If you used AI tools (e.g. ChatGPT, Copilot, Claude), walk through the actual prompts and conversations you had.

Use of AI Tools

AI can be a powerful assistant, but keep in mind it's not always accurate. Just like collaborating with a teammate, it's your responsibility to validate the output, ensure it's correct, and be able to explain it in a clear, human-readable way. AI may help get you 90% of the way there, but the remaining 10% often requires careful navigation, critical thinking, and troubleshooting to reach a reliable solution. If you use AI during the exercise, please make sure to share how you used it, eg: the kinds of questions you asked and how you applied or adapted the responses—alongside your solution.

Our goal is to assess not just the final answer, but also how you approach problem-solving, how you leverage available tools responsibly, and how you ensure quality in your work. Think of this as simulating how you would tackle a real-world problem in your day-to-day job.

Question 1: Build a webhook event streaming integration to send events to NebulaReach

Background

Amplitude is a leading digital analytics platform that helps teams understand user behavior, build better products, and drive business outcomes. With powerful analytics and customer journey insights, it empowers product and growth teams to make data-driven decisions. In addition to its core analytics capabilities, Amplitude also connects seamlessly with a wide range of tools and platforms, enabling customers to send event data into or out of Amplitude. This is made possible through Amplitude Activation, which includes over 100 pre-built integrations with top platforms like Google Ads, Meta Ads, HubSpot, and many more.

However, customers often use a diverse set of tools in their tech stack and may require integrations beyond those available out of the box. To support these broader needs, Amplitude provides a flexible webhook integration feature that allows users to define custom destinations by specifying a URL, request headers, and payload structure.

We have an urgent deal commitment with our customer DataMonster, who wants to integrate Amplitude with NebulaReach—a cutting-edge marketing automation platform used by AI-first startups to personalize multi-channel user journeys in real time.

Since NebulaReach is not yet a built-in destination in Amplitude, we need to deliver a quick proof of concept (POC) to demonstrate that Amplitude can stream events to NebulaReach using a webhook-based integration. You are asked to build a transformation layer that takes in an Amplitude event and converts it into the format expected by NebulaReach, based on the following rules.

Transformation Rules

- `user_id` in NebulaReach = **SHA-256 hash** of the `email` field from `user_properties` in the Amplitude event
- `phone_number` in NebulaReach
 - Take `phone_number` from `user_properties`
 - Remove all **non-digit** characters
 - Hash the cleaned number using **SHA-256**
- `external_id` in NebulaReach = **SHA-256 hash** of the `external_id` field from `user_properties`
- `event_time` in NebulaReach = **Epoch time (in seconds)** derived from the `event_time` field in the Amplitude event
- `event` in NebulaReach = the `event_type` from the Amplitude event

- **event_properties** = Copied **as-is** from the Amplitude event
- **user_properties** in the final payload should **only include** the hashed **phone_number** and **external_id**

💡 **Note:** Event properties may be unlimited in both Amplitude and NebulaReach. Your transformation should preserve **all** properties without filtering or modification.

Example Amplitude event payload:

```
{
  "event_type": "Purchase",
  "user_id": "03b3fc3261dfb2e58c5dc966071b7aae10f98228e3668fed17b946129d9c219a",
  "event_time": "2025-03-21 22:01:40.158000",
  "user_properties": {
    "email": "testing@amplitude.com",
    "phone_number": "(604) 250-1234",
    "external_id": "id_6042501234"
  },
  "event_properties": {
    "location": "Vancouver",
    "source": "hubspot",
    "type": "jacket",
    "brand": "lululemon",
    "price": 75
  }
}
```

Example NebulaReach Event payload

```
{
  "event": "Purchase",
  "user_id": "27789f01caab55b9b13b5a8f42a0a0738a68e27316561ba8325ee1d19963c767",
  "event_time": "1742594794",
  "user_properties": {
    "phone_number":
"def95eae897c3e73b952fb0f564fb61d8c8a8b85e5f1a4d7788eb129ea215e51",
    "external_id":
"b243a3ecb1634ba50a3b2f83c77d90af93f1063bc30047e3c754061186455e0f"
  },
  "event_properties": {
    "location": "Vancouver",
```

```
"source": "hubspot",
"type": "jacket",
"brand": "lululemon",
"price": 75
}
}
```

Question

1. Create a Webhook Streaming Destination in Amplitude

- Log into your Amplitude organization.
- In the **left navigation bar**, click **“Data”**, then in the secondary nav bar, click **“Catalog.”**
- At the top, select the **“Destinations”** tab.
- Search for **“Webhook”**, then under **“Messaging Destinations”**, click **“Webhook.”**
- Click **“Add Destination”**, and give your webhook sync a meaningful name.

 *Tip:* If you close the page and want to revisit it later, go to **Data > Destinations**. You'll find your webhook sync listed there.

2. Set Up a Temporary Webhook URL

- Go to <https://webhook.site/> to generate a URL where you can inspect incoming webhook requests.
 - Copy the generated URL and use it as the destination URL in your Amplitude integration.
 - Once the integration is set up, this will let you test the transformed events and compare them against the expected output format.
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3. Define a Custom Payload Format

- In the **“Define Event Payload”** section of your Amplitude destination setup, select **“Customized Payload.”**
 - You'll now be able to define your own transformation logic.
 - Use **FreeMarker Template Language (FTL)**—Amplitude's templating engine—to write the transformation rules based on the mapping provided earlier.
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4. Test Your Integration

- After setting up the transformation, click “**Test Connection**” in the top-right corner.
- Copy and paste the provided sample Amplitude event as your test input.
- Click “**Send Event.**” If the configuration is correct, you’ll see the transformed payload appear on your webhook.site page.
- Compare the output with the expected NebulaReach event format to ensure everything is working correctly.
- Once you finish, click “**Save Connection.**”

Evaluation Points

Your solution will be evaluated based on the following:

- Events are successfully sent to webhook.site
- `event_time` is correctly converted to epoch time (in seconds)
- `user_id` is correctly hashed from the email
- `phone_number` and `external_id` are correctly cleaned and hashed
- All `event_properties` are fully preserved in the output

Useful Tips

If you're unfamiliar with FreeMarker, you can experiment with it using the [FreeMarker Online Playground](https://freemarker.org/docs/mrCh01.html). In this integration, the Data Model represents the incoming event payload from Amplitude, and the Template defines how you want to structure the outgoing webhook payload.

- For example, if your data model is:
`{"user_properties": {"id": "test_id"}}`
- And your template is:
`${input.user_properties.id}`
- The output will be:
`test_id`

Online FreeMarker Template Tester

Template (Apache FreeMarker 2.3.34)

`${input.user_properties.id}`

Data model ([show examples](#))

```
input = {  
  "user_properties": {  
    "id": "test_id"  
  }  
}
```

Output format ([?](#))

undefined ▾

Locale

English (United[...]; en_US ▾

Time zone

America/Los_Angeles ▾

Tag syntax ([?](#))

angleBracket, like <#...>, <@...> ▾

Interpolation syntax ([?](#))

legacy, like \${...}, #{...} ▾

Evaluate

Ctrl+Enter in input fields will submit this form too

Result

test_id

Question 2: Ingesting Events from HubSpot into Amplitude

Background

Now that you've seen how Amplitude streams events to third-party destinations, let's look at the other side of Amplitude Activation—event ingestion from external sources. Amplitude supports pulling events from various platforms, transforming them, and ingesting them into Amplitude using the HTTP V2 API. One real-world example is HubSpot, one of our top partners. In the past, Amplitude built an integration to ingest email engagement events from HubSpot.

Question

1. Pulling Events from HubSpot

Using the HubSpot documentation below:

- Event Reference:
<https://developers.hubspot.com/docs/guides/api/analytics-and-events/email-analytics#event-references>
- API Reference:
<https://developers.hubspot.com/docs/reference/api/analytics-and-events/email-analytics>

Write a small script (in any language of your choice) that pulls events from HubSpot's email analytics API. (Note: It's okay if the actual API call returns an empty array. As long as the script gets a 200 OK response, it will be reviewed offline by an engineer.)

Please use the following testing credentials:

- Client ID: '6811cb3d-e012-495a-8295-d5f63bf87a19'
- Client Secret: 'db126ef8-3ef8-48b1-b804-38ceabbb1fae'
- Refresh Token: '8ebf04ee-4c0e-4be4-adca-644112dceb49'

2. Sending Events to Amplitude

Assume the HubSpot API response is in the format shown below.

```
{  
  "appId": 123456,  
  "portalId": 7890123,  
  "recipient": "recipient_testing@amplitude.com",  
  "type": "DROPPED",  
  "timestamp": 1742849711,  
  "hs_email_campaign_id": 4567890,  
  "hs_email_id": 1234567,
```



```
"sendId": "abc123-def456-ghi789",
"smtpId": "202503240001.abc123@example.com",
"dropReason": "VALIDATION_FAILED",
"dropMessage": "Email not allowed to be sent to the specified recipient in a
sandbox portal",
"messageIdHash": "abcdef1234567890",
"sentBy": "sender_testing@amplitude.com"
}
```

Your task is to write a script that takes this HubSpot email activity event and transforms it into the format required by Amplitude's HTTP V2 API. The event should be mapped to a corresponding Amplitude event, according to Amplitude HubSpot Integration documentation. The script should extract the relevant fields (such as eventType, recipient, and timestamp), reformat them to match Amplitude's ingestion schema via HTTP V2 API.

Using the Amplitude documentation below:

- Amplitude HTTP v2 API: <https://amplitude.com/docs/apis/analytics/http-v2>
- Amplitude's HubSpot ingestion integration: <https://amplitude.com/docs/data/source-catalog/hubspot>

Evaluation Points

Your solution will be evaluated based on the following:

- Successfully obtains an access token from the provided refresh token.
- Makes the correct API call to retrieve email activity events from HubSpot.
- Accurately transforms the events into the format expected by Amplitude's HTTP V2 API.
- Sends the transformed events to Amplitude, and verifies they appear in the "Live Events" view within the Amplitude project.